

PIETERMARITZBURG, South Africa

WMO: 685810

Lat: 29.6278S

Lon: 30.4028E

Elev: 673

StdP: 93.50

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	5.9	-3.3	3.1	17.4	-1.5	3.6	15.2	5.5	17.3	4.7	20.7	0.4	240	0.284

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.9	34.1	21.1	32.3	21.2	30.6	21.2	24.7	29.7	23.8	28.5	23.1	27.5	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	19.6	27.4	22.5	18.7	26.3	21.8	17.9	25.4	79.0	29.6	75.4	28.6	72.4	27.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.3	4.4	3.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			1.3	26.2	1.9	1.3	0.0	27.1	-1.1	27.9	-2.2	28.6	-3.5	29.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	22.9	23.4	22.7	20.3	17.7	14.9	14.8	17.0	18.5	19.2	20.3	21.7
	DBStd	4.22	2.89	2.81	2.94	3.05	2.66	2.33	2.80	3.28	3.83	3.67	3.55	3.21
	HDD10.0	4	0	0	0	0	0	1	2	1	1	0	0	0
	HDD18.3	457	2	1	3	16	44	106	114	64	44	35	19	8
	CDD10.0	3440	398	375	394	309	238	147	150	218	254	285	308	364
	CDD18.3	854	143	143	140	74	23	2	4	23	48	62	78	113
	CDH23.3	5451	854	815	798	399	176	49	64	222	396	466	518	694
CDH26.7	1838	302	276	278	124	34	4	6	65	148	172	189	241	
Wind	WSAvg	1.3	1.5	1.4	1.3	1.1	1.0	1.0	1.1	1.3	1.5	1.6	1.6	1.6
Precipitation	PrecAvg	871	138	118	110	49	20	13	14	26	52	82	104	119
	PrecMax	1286	299	314	298	155	98	62	108	87	444	178	261	293
	PrecMin	585	47	26	21	5	0	0	0	0	6	23	36	32
	PrecStd	167	59	65	66	33	20	16	23	23	74	37	47	60
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	35.1	34.4	33.3	30.1	27.0	27.3	31.4	34.7	35.0	35.2	34.6
		MCWB	22.1	22.3	21.5	19.7	16.0	14.0	13.2	15.3	17.3	19.1	20.5	22.0
	2%	DB	32.3	32.1	31.8	29.8	27.4	24.6	25.2	28.3	30.7	31.3	31.6	31.6
		MCWB	22.9	22.9	22.1	19.5	16.3	13.3	13.1	14.8	17.1	19.4	21.3	22.6
	5%	DB	29.9	29.9	29.7	27.6	25.2	22.6	23.0	25.7	27.7	28.5	28.9	29.3
		MCWB	22.8	23.0	22.0	19.1	16.0	12.8	13.0	14.5	16.9	19.4	21.0	22.5
	10%	DB	27.8	28.0	27.7	25.3	23.2	20.6	20.6	23.1	25.0	25.7	26.3	27.3
		MCWB	22.4	22.5	21.6	18.8	15.8	12.5	12.4	14.4	16.6	18.6	20.3	21.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.7	25.0	22.0	19.2	16.5	16.1	18.7	20.5	22.7	24.4	25.5
		MCDB	30.5	30.7	30.0	28.3	25.1	21.4	23.4	24.3	28.0	29.7	30.7	30.8
	2%	WB	24.2	24.4	23.6	21.0	18.2	15.4	14.9	17.3	19.1	21.1	22.8	24.2
		MCDB	29.1	29.1	28.3	26.3	23.1	19.6	20.9	22.9	26.1	27.1	28.2	28.8
	5%	WB	23.4	23.6	22.7	20.1	17.3	14.4	14.0	16.2	18.1	20.0	21.7	23.1
		MCDB	27.9	27.9	27.2	24.6	21.7	19.3	19.2	21.9	24.0	25.7	26.3	27.4
	10%	WB	22.5	22.8	21.9	19.4	16.6	13.5	13.2	15.2	17.2	19.1	20.8	22.2
		MCDB	26.8	26.6	26.2	23.4	20.9	18.7	18.4	20.9	22.5	23.8	24.9	26.0
Mean Daily Temperature Range	5% DB	MDBR	8.7	8.9	9.5	10.6	12.8	14.0	13.7	12.7	11.4	10.0	9.0	8.6
		MCDBR	13.1	12.3	13.2	14.3	15.8	16.8	17.0	17.4	17.1	16.2	14.6	13.0
		MCWBR	5.0	4.6	4.7	4.8	5.8	6.7	7.1	6.0	5.8	6.0	5.6	5.6
	5% WB	MCDBR	11.2	10.7	11.2	11.5	12.4	13.2	14.0	13.0	13.7	12.8	11.6	11.5
		MCWBR	5.1	4.8	4.9	4.6	5.6	6.5	7.1	5.5	5.8	5.7	5.7	5.8
Clear-Sky Solar Irradiance	taub	0.422	0.416	0.404	0.369	0.341	0.323	0.333	0.394	0.443	0.459	0.432	0.426	
	taud	2.314	2.332	2.346	2.400	2.440	2.465	2.426	2.258	2.144	2.148	2.254	2.290	
	Ebn at Noon	921	908	885	866	844	839	843	825	832	858	906	920	
	Edh at Noon	138	132	123	105	91	84	90	119	148	157	146	143	
All-Sky Solar Radiation	RadAvg	5.72	5.65	5.02	4.19	3.64	3.27	3.51	4.07	4.68	4.95	5.39	5.67	
	RadStd	0.41	0.45	0.26	0.27	0.20	0.17	0.21	0.24	0.29	0.48	0.46	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page